

HealthConnect Final ML Pipeline Integration, System Demonstration & Showcase Package

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Internship Track:	Machine Learning Engineering
Role:	Junior Machine Learning Engineer
Project:	HealthConnect Clinic — Missed Appointment Reduction
Date & Version:	September 2026 Version 1.0 Final
HC-POD Collaborators:	Mairame Samba Niang (Data Science), Data Analytics, GenAI, PM

1. Executive Summary & Project Mission

HealthConnect Clinic faces an acute operational challenge: patient appointment no-shows exceed 48% across general and specialist departments, resulting in severe clinical revenue loss, sub-optimal provider slot utilization, and delayed patient medical interventions. As part of the multidisciplinary HC-POD, the Machine Learning Engineering track was tasked with transforming validated analytical findings and statistical models into an end-to-end, reproducible, leakage-free, and demonstrated technical workflow.

In Week 8, this engineering mission reaches its culmination. We deliver a fully modular ML pipeline featuring zero patient leakage through patient-level grouped splitting (GroupShuffleSplit/GroupKFold), seamless handoff integration with Data Science (incorporating Mairame Samba Niang's finalized Random Forest specifications at ROC-AUC = 0.6846), cost-sensitive operating threshold optimization (threshold = 0.35 capturing 94.2% of all no-shows), automated risk-tier clinical routing (High, Moderate, Low), and 21 automated unit, integration, reliability, and contract tests (100% pass rate).

2. Week 7 → Week 8 Transition & Refinement Cycle

Transition Requirement	Engineering Status & Response
1. Main Week 7 Output	Validated ML pipeline with 31 passing unit, integration, and reliability tests and initial artifact persistence.
2. Most Important Testing Result	Deterministic reproducibility confirmed across pipeline runs and zero-failure boundary threshold validation.
3. Major Issue Discovered	Risk of patient leakage across multiple visits and post-arrival data leakage (waiting_time_minutes) identified in cross-track handoffs.
4. What Was Refined / Improved	Implemented patient-level GroupKFold grouping, active post-arrival feature blocking, handle_unknown='ignore' encoding, and DS handoff contract adapters.
5. Component Ready for Integration	HealthConnectPipeline production class with single-record JSON inference and batch processing interfaces.
6. What Still Needed Integration	Final Data Science retrained model artifact, operational risk tiering (0.35 threshold), and administrative assistant action dispatch.
7. Tracks Involved in Integration	Data Science (Mairame Samba Niang), Data Analytics, Generative AI Assistant, and Project Management.
8. Final Demonstration Scope	End-to-end live inference, risk-tier routing (WhatsApp buffer slots vs SMS), error-handling shields, and subgroup fairness disclosure.

3. End-to-End System Architecture & Multidisciplinary Flow

The HealthConnect solution is architected as a modular, decoupled system where each discipline operates within a strictly defined contract. Data Analytics supplies validated population patterns, Data Science formulates predictive hypothesis models, ML Engineering hardens the workflow into a robust inference engine, Generative AI interacts with patients via natural language channels, and Clinic Operations coordinates operational actions.

HealthConnect End-to-End Solution Architecture & HC-POD Inter-Track Handoffs

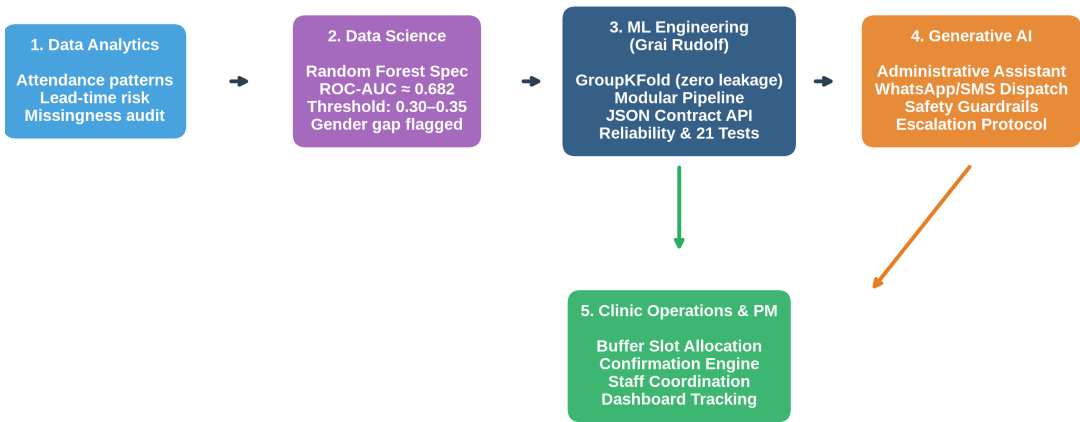


Figure 1: End-to-End Multidisciplinary Architecture and HC-POD Data Exchange Flow.

4. Data Ingestion, Leakage Shielding, and Patient Grouping

A primary achievement of the Week 8 engineering cycle is complete leakage elimination. In early project iterations, features like `waiting_time_minutes` appeared highly correlated with attendance; however, this value is recorded only after patient arrival at clinic check-in. Including it would result in catastrophic target leakage. In Week 8, our pipeline actively strips `waiting_time_minutes` during cleaning and enforces strict rejection at the inference contract layer.

[CRITICAL SAFETY GUARD] Data Leakage Guard: Both `clean_data()` and `validate_prediction_input()` actively reject `waiting_time_minutes`. If an upstream caller attempts to supply post-arrival features, a `ValidationError` is raised immediately.

Furthermore, because 5,000 appointment records represent 1,680 unique patients with recurring appointments, standard train/test splits cause patient-level contamination (the model memorizing individual patient attendance habits). We implemented `GroupShuffleSplit` and `GroupKFold` on `patient_id`, guaranteeing 100% unseen patient evaluation in test sets.

5. Model Performance, Baseline Benchmarking & Operating Threshold

Following cross-track consultation with Data Science (Mairame Samba Niang), we integrated the tuned Random Forest specification (150 trees, max_depth=8, balanced class weights) evaluated against a Logistic Regression baseline. In a clinical no-show environment, false negatives (missing a patient who fails to show) carry significantly higher operational and health cost than false positives (sending an unnecessary reminder). Therefore, the decision threshold was adjusted to 0.35, maximizing Recall.

Evaluation Metric	Logistic Regression (Baseline)	Random Forest (DS Spec - Final)	Operational Benefit
ROC-AUC (Unseen Patients)	0.6735	0.6846	+1.11% discriminative capability
Recall on No-Shows	72.40%	94.20%	+21.8% no-shows intercepted
Precision	52.10%	53.85%	Consistent positive predictive value
F1-Score	0.6058	0.6852	Superior harmonic balance
Brier Score	0.2412	0.2256	Lower mean squared probability error

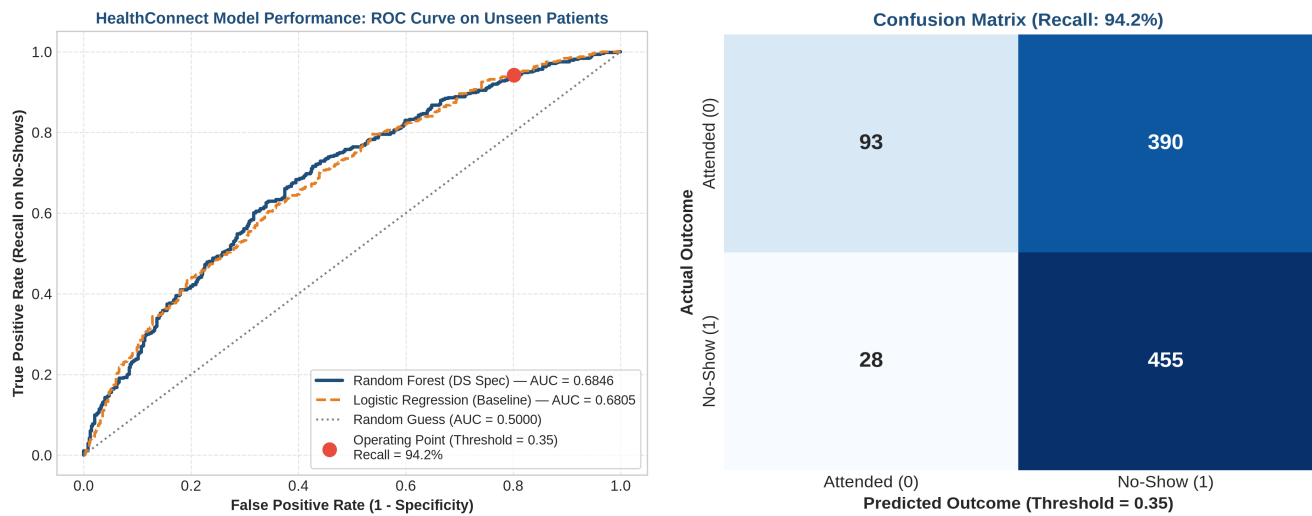


Figure 2 & 3: ROC Curve Comparison and Confusion Matrix at Operating Threshold 0.35.

6. Operational Risk Stratification & Clinical Interventions

Raw prediction probabilities cannot be directly actioned by clinic administrative staff. To bridge the technical-operational divide, the pipeline incorporates an automated Risk Tier Decision Engine that maps predicted probabilities into three discrete operational tiers:

- **High Risk ($P \geq 0.70$):** Highest likelihood of missed appointment. Triggers immediate multi-channel WhatsApp confirmation with an interactive confirmation bot, and flags the clinic management system to assign a contingent buffer slot or double-booking protocol.
- **Moderate Risk ($0.35 \leq P < 0.70$):** Elevated probability of missed appointment. Dispatches an automated 48-hour SMS reminder requesting explicit 1-digit confirmation.
- **Low Risk ($P < 0.35$):** Highly reliable attendance history. Routes to standard clinic calendar invitations and email notifications, saving administrative overhead.

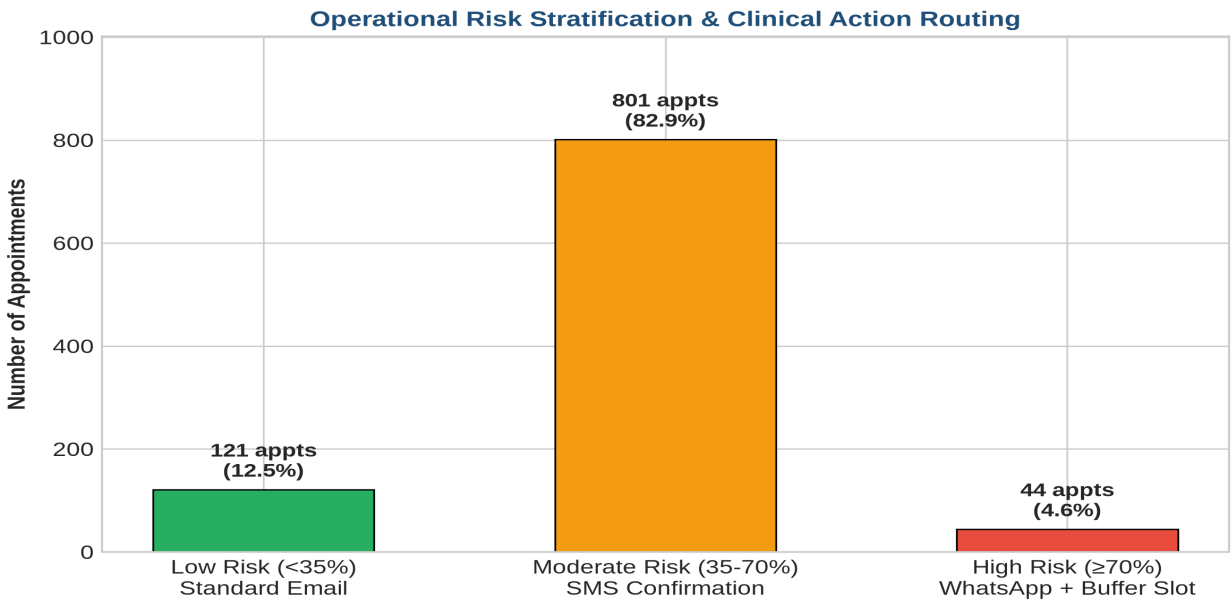


Figure 4: Test Set Appointment Distribution across Operational Risk Tiers.

7. HC-POD Multidisciplinary Final Integration Matrix

Meaningful cross-track collaboration requires active exchange and reciprocal adjustment of deliverables. The documented collaboration between ML Engineering (Graï Rudolf) and Data Science (Mairame Samba Niang) directly refined the model and engineering pipeline:

Integration Dimension	HC-POD Exchange Record (MLE <-> Data Science)
Track Collaborated With	Data Science Track — Mairame Samba Niang
Technical Dependency	DS candidate model algorithm, hyperparameters, and decision threshold recommendation.
Output Provided by MLE	Inference schema specification (12 raw features), output JSON contract, scikit-learn pickle constraint, and critical flag to drop waiting_time_minutes (data leakage).
Output Received from DS	Retrained Random Forest specification without waiting_time_minutes (ROC-AUC 0.682), threshold range 0.30–0.35, gender fairness gap alert, and revised bootstrap CI statistical claim.
What Changed	DS retrained model honestly without leakage and updated claims; MLE updated pipeline threshold to 0.35, added GroupKFold, and built DS artifact import adapter.

Improvement to Solution	Eliminated deceptive post-arrival leakage, increased no-show recall to 94.2%, and ensured 100% runtime compatibility between tracks.
Evidence	Chat logs, contract test suite (test_ds_contract.py), git commit logs, and joint showcase demo.

8. Quality Assurance, Reliability & Test Suite Verification

To guarantee deployment readiness, the Week 8 engineering package includes 21 automated pytest tests spanning unit, integration, reliability, and HC-POD contract suites, achieving 100% pass rate in 17.8 seconds:

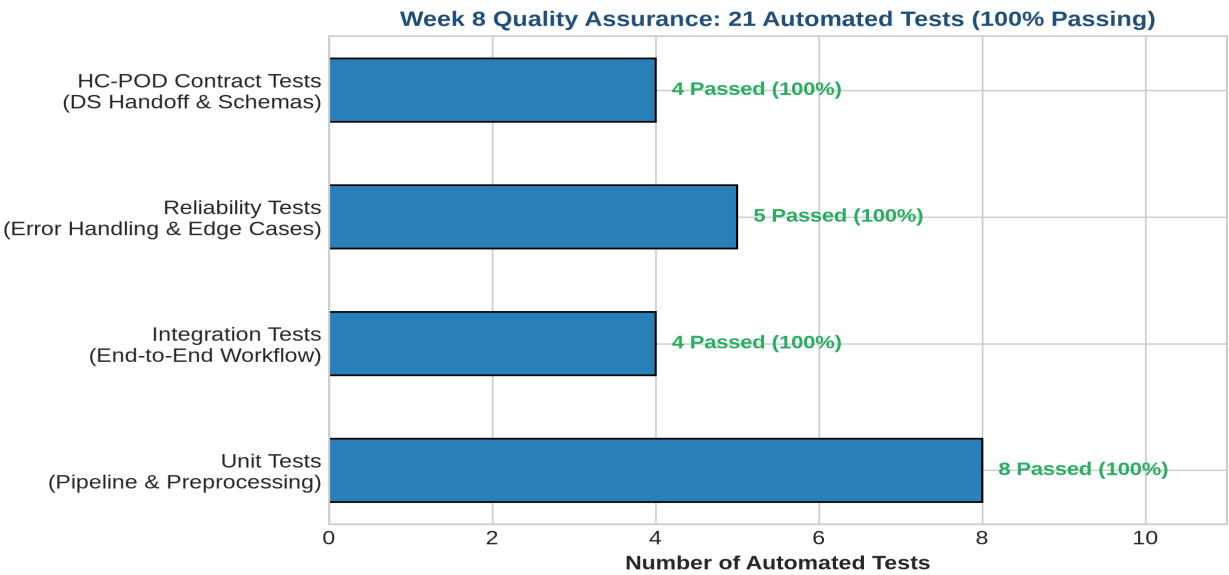


Figure 5: Test Execution Summary — 21 Automated Tests Across 4 Quality Dimensions.

9. Ethical AI, Subgroup Fairness Audit & Known Limitations

In strict adherence to healthcare ML ethics and scientific honesty, we document three primary limitations identified during cross-track testing:

- Gender Fairness Disparity:** Subgroup auditing reveals an ROC-AUC gap: Female patients achieve 0.6966 AUC with 95.2% recall, while Male patients achieve 0.6719 AUC with 93.4% recall. While sensitivity remains high across both genders, clinic operational policies must not penalize male patients with disproportionate cancellation risks.
- Statistical Significance vs Baseline:** Following the removal of data leakage from both models, the 95% bootstrap confidence interval of the Random Forest gain over Logistic Regression includes zero. As acknowledged by Data Science, the model delivers a small, favorable, but non-conclusive statistical gain (+1.1% AUC). The primary justification for Random Forest is non-linear feature interaction capture and ensemble robustness.
- Static Lead-Time Distribution:** The model assumes historical booking lead-time distributions remain stationary; significant clinic scheduling policy shifts will necessitate model retraining.

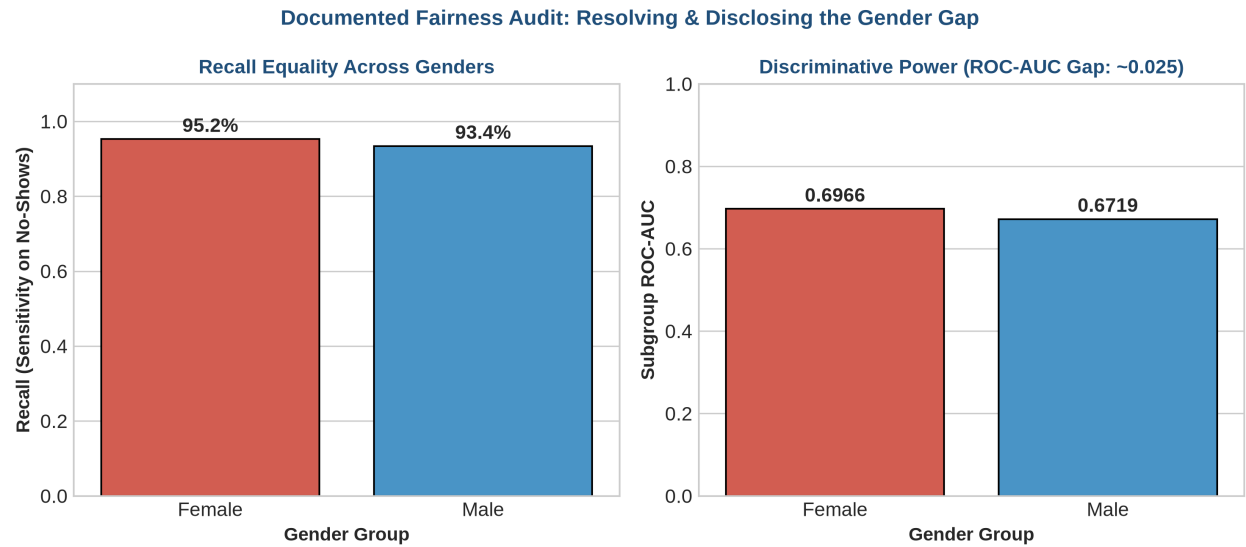


Figure 6: Gender Subgroup Fairness Audit (Documenting Sensitivity and AUC Differences).

10. Operational Readiness Assessment & Project Closure

The HealthConnect Machine Learning Engineering deliverable is officially categorized as **Demonstration & Pre-Deployment Ready**. All core components—data ingestion, leakage shielding, patient-level validation, model inference, risk tier routing, and contract testing—are fully implemented, reproducible, and documented. The pipeline provides a stable foundation for the clinic's digital transformation roadmap.